

FERHAN BURAK ÖZKAN

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer specializing in frontend and mobile development across **React Native, Expo, TypeScript, Next.js, and Flutter**. Proven track record of shipping end-to-end digital products from concept to production, including an active **live, production-approved GPS navigation app on Google Play (TrackSnap)** and custom-domain web platforms. Winner of **1st Place at Çankaya University's 18th R&D Market** for a TÜBİTAK-backed healthcare IoT solution, with hands-on experience in open-source LLM fine-tuning and computer vision.

TECHNICAL SKILLS

Mobile: React Native, Expo, Expo Router, TypeScript, React 19, Flutter, Dart, Google Maps SDK, Expo Location, Task Manager, Push Notifications, SecureStore, BLE

Frontend & Web: React, Next.js, Vite, TypeScript, JavaScript (ES6+), Tailwind CSS, HTML5, CSS3, Responsive Layouts, Server-Side Rendering (SSR)

Network & Cybersecurity: Firewalls, IDS / IPS, OSI Model Layers, Network Traffic Analysis, Threat Mitigation, Packet Inspection, Network Security Architectures

AI & Machine Learning: Python, LLM Fine-Tuning (Unsloth, LoRA, 4-bit Quantization), Instruction Tuning, CNN (Convolutional Neural Networks), Computer Vision, PyTorch, Google Colab Pro

Backend & Cloud: Firebase (Auth, Firestore, Realtime Database, Cloud Functions, Cloud Storage, App Check, Security Rules, Emulator Suite), REST APIs, Node.js, Express, Vercel

Testing & DevOps: Vitest, React Testing Library, Firebase Rules Testing, ESLint, EAS Build, Google Play Console (Closed Testing & AAB Releases), Git, GitHub, Postman, Docker, Selenium

Databases & Design: PostgreSQL, MySQL, Figma (UI Design & Prototyping)

WORK EXPERIENCE

CBERNET , Cybersecurity & Network Engineer Intern

October 2025 - September 2026

- Gained 1 year of hands-on experience in enterprise network defense, analyzing traffic across the 7 layers of the OSI model to identify protocol anomalies and security breaches.
- Configured and monitored Next-Generation Firewalls alongside Intrusion Detection and Prevention Systems (IDS / IPS) to filter malicious packets and enforce perimeter security policies.
- Participated in vulnerability assessments, internal network audits, and automated log analysis to maintain system hardening and compliance standards.

Nokia , Software Engineer Intern

July 2025 - September 2025

- Built an internal AI chatbot to help engineering teams quickly search and navigate technical network documentation.
- Wrote automated end-to-end regression and UI test suites using Selenium to catch visual and functional regressions before releases.
- Worked in an Agile environment with daily standups, bi-weekly sprint planning, and regular peer code reviews.

BOTAŞ , Software Engineer Intern (Mobile Development)

July 2023 - August 2023

- Developed a mobile inventory management app in **Flutter (Dart)** to digitize daily warehouse tracking.
- Integrated on-device machine learning models for real-time barcode and item recognition, cutting down manual inventory logs.
- Implemented background data synchronization to ensure offline entries update correctly against company databases.

EDUCATION

Çankaya University , Bachelor of Science in Software Engineering (English)

2020 - 2025

Ankara, Türkiye

LANGUAGES

Turkish: Native

English: Full Professional (iTEP Level 4)

German: Elementary

FEATURED PROJECTS

TrackSnap - GPS Driving & Route Logging App

2026 - Present

Creator & Lead Mobile Developer

Published on Google Play • Production Track Approved • iOS App Store Review in Progress

- Architected a mobile app in **React Native, Expo Router, and TypeScript (React 19)** paired with a **React, Vite, and Tailwind CSS** admin portal, delivering background GPS tracking and live map views via **Google Maps SDK, Expo Location, and Task Manager**.
- Integrated backend services with **Firebase (Auth, Firestore, Realtime DB, Cloud Functions)**, secured via **Firebase App Check (Play Integrity)** and validated with automated **Vitest, React Testing Library**, and local **Firebase Emulator Suite** tests.
- Directed the complete Google Play release pipeline: fulfilled closed testing with **20+ active testers**, configured automated EAS production AAB builds, and resolved store compliance.

Tech Stack: React Native, Expo Router, TypeScript, React 19, Firebase (Functions, App Check, Firestore), Google Maps SDK, Vite, Vitest, EAS Build

BALSA Dekor - Furniture Catalog & Admin Web App

January 2026 - Present

Full Stack Frontend Developer

Live Website: balsadekor.com

- Engineered and deployed a production catalog showcase for **50+ custom furniture products** using **Next.js, React, Tailwind CSS**, and **Framer Motion** for sub-second page transitions.
- Constructed a secure admin panel using **Firebase Auth, Firestore, and Storage** to manage inventory, update product listings, track sales, and process inquiries.
- Configured granular Firestore security rules, connected form pipelines via **FormSubmit**, and managed custom domain DNS and CI/CD on **Vercel**.

Tech Stack: Next.js, React, JavaScript, Tailwind CSS, Firebase (Auth, Firestore, Storage, Security Rules), Framer Motion, Lucide React, Vercel

ScoliSense - Real-Time IoT Healthcare App

September 2024 - Present

Lead Mobile & Frontend Developer

1st Place Winner - Çankaya Univ. 18th R&D Market • TÜBİTAK 2209-A Funded • Patent Pending (Preliminary Examination Passed) • [► Live Demo Video \(YouTube\)](#)

- Built a cross-platform **Flutter (Dart)** mobile app that streams live pressure data from an Arduino-based smart scoliosis brace over **Bluetooth Low Energy (BLE)**.
- Developed an **Angular** clinician portal synchronized via **Firebase Realtime Database** and Auth to track patient brace wear time, compliance graphs, and sensor history.
- Collaborated directly with medical specialists at SBU Gaziler Hospital on clinical validation, and co-authored the project's patent application, which successfully passed preliminary examination (Grant Pending).

Tech Stack: Flutter, Dart, Angular, Firebase, Arduino BLE, Vercel

Applied AI & Machine Learning Engineering

March 2025 - June 2025

AI & Deep Learning Developer

Code Repositories: [Turkish LLM Fine-Tuning](#) • [Traffic Sign CNN](#)

- Turkish LLM Fine-Tuning:** Fine-tuned open-source LLMs on curated instruction datasets via **Unsloth**, applying **LoRA and 4-bit quantization** on Google Colab Pro for enhanced Turkish reasoning and generation.
- Traffic Sign Image Classification:** Engineered a Convolutional Neural Network (CNN) in **PyTorch** to classify 42 traffic sign categories from the GTSRB dataset, achieving a **98% test accuracy**.

Tech Stack: Python, Unsloth, LLM Fine-Tuning (4-bit Quantization, LoRA), CNN, Computer Vision, Google Colab Pro, PyTorch

HONORS & CERTIFICATIONS

- OpenAI Academy:** Apply AI at Work Pathway Certificate — Completed all 3 courses with assessments scored $\geq 80\%$ (Credential ID: 194332281)
- 1st Place Award:** Çankaya University 18th R&D Project Market (ScoliSense Project)
- Hackathon Certificate:** Techcareer.net Hackathon Participant & Project Finalist (Credential ID: 63298683643333)
- iTEP English Proficiency Exam:** Level 4 - Full Professional Proficiency (Credential ID: 15763B63CNV)